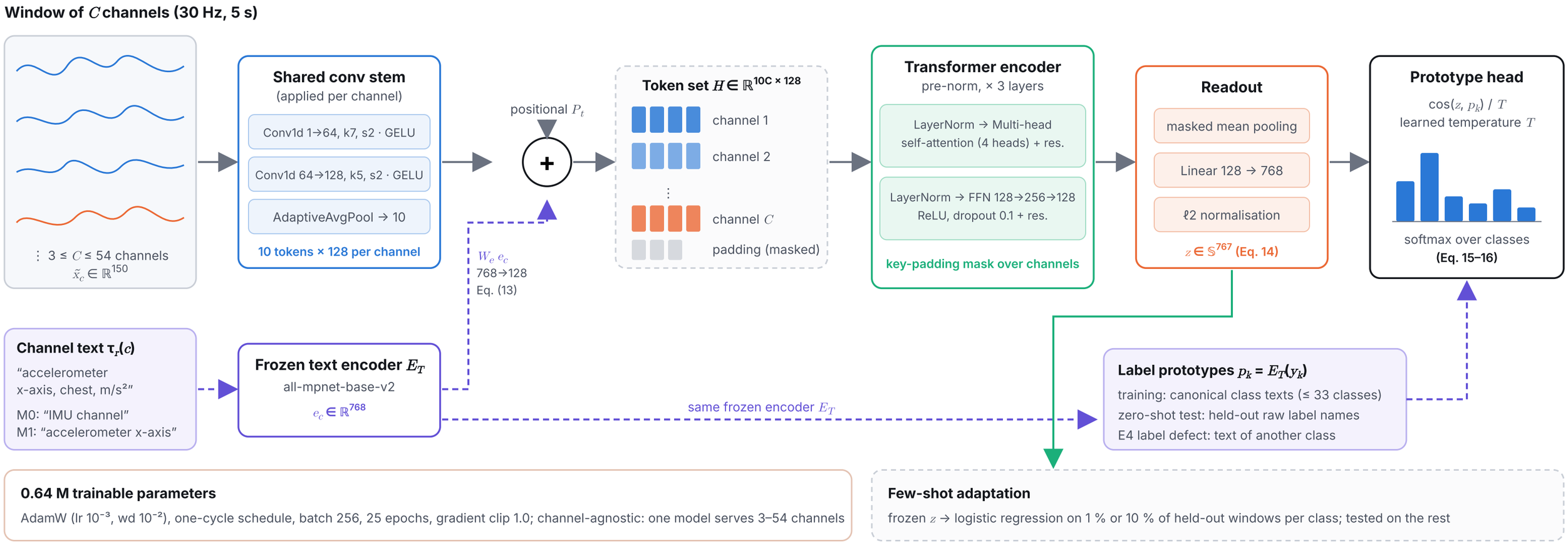

Window of C channels (30 Hz, 5 s)
⋮ 3 ≤ C ≤ 54 channels
$\tilde{x}_c \in \mathbb{R}^{150}$
Shared conv stem
(applied per channel)
Conv1d 1→64, k7, s2 · GELU
Conv1d 64→128, k5, s2 · GELU
AdaptiveAvgPool → 10
10 tokens × 128 per channel
positional P_t
+
Token set $H \in \mathbb{R}^{10C \times 128}$
channel 1
channel 2
⋮
channel C
padding (masked)
$W_e\, e_c$
768→128
Eq. (13)
Transformer encoder
pre-norm, × 3 layers
LayerNorm → Multi-head self-attention (4 heads) + res.
LayerNorm → FFN 128→256→128 ReLU, dropout 0.1 + res.
key-padding mask over channels
Readout
masked mean pooling
Linear 128 → 768
ℓ2 normalisation
$z \in \mathbb{S}^{767}$ (Eq. 14)
Prototype head
$\cos(z, p_k) / T$
learned temperature T
softmax over classes
(Eq. 15–16)
Channel text $\tau_r(c)$
"accelerometer x-axis, chest, m/s²"
M0: "IMU channel"
M1: "accelerometer x-axis"
Frozen text encoder E_T
all-mpnet-base-v2
$e_c \in \mathbb{R}^{768}$
same frozen encoder E_T
Label prototypes $p_k = E_T(y_k)$
training: canonical class texts (≤ 33 classes)
zero-shot test: held-out raw label names
E4 label defect: text of another class
0.64 M trainable parameters
AdamW (lr 10^{-3}, wd 10^{-2}), one-cycle schedule, batch 256, 25 epochs, gradient clip 1.0; channel-agnostic: one model serves 3–54 channels
Few-shot adaptation
frozen z → logistic regression on 1 % or 10 % of held-out windows per class; tested on the rest